SENTIMENT ANALYSIS OF DISNEYLAND REVIEWS

Izabela Kosiec 441755 Katarzyna Pełka 432453

Agenda

- Dataset overview
- Text preprocessing
- Dictionary based methods
- Supervised sentiment analysis method
- Conclusions

Data overview

Dataset:

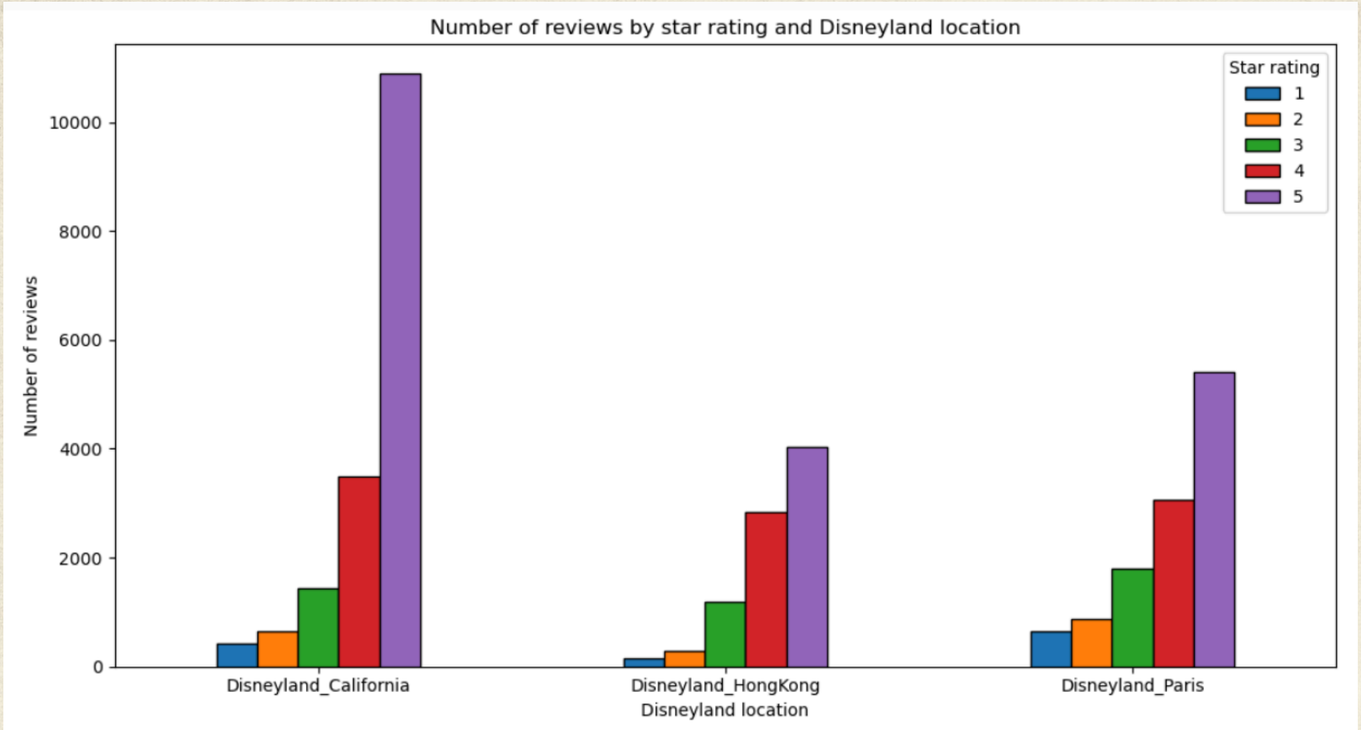
- Review_ID
- Rating – integer rating (1–5)
- Year_Month
- Reviewer_Location
- Review_Text
- Branch: Disneyland_HongKong, Disneyland_Paris, Disneyland_California

rating_bin Branch	negative (1-2)	neutral (3)	positive (4-5)
Disneyland_California	1056	1447	14382
Disneyland_HongKong	411	1181	6864
Disneyland_Paris	1518	1805	8466

Rating bin

- Positive (4–5) 33921
- Neutral (3) 5109
- Negative (1–2) 3626

Total reviews: 42 656



Wordcloud of reviews

park
ride
Disneyland
Disney
parade
even
place
will
good
lot
show
day
queue
line
fun
much
time
need
one
still
went
take
food
people
visit
see
go
kid
staff
family
really
great
wait
got
attraction
experience
fast pass
Space Mountain
California Adventure
Disneyland Paris
money
thought
visited
enjoy
Main Street
wanted
know
found
think
going
open
always
hour
minute
ticket
amazing
children
Paris
Disney park
well
worth
expected
never
made
took
end
back
bad
around
best
magical
hotel
trip
first
price
firework
restaurant
make
enough
everyone
thing
night
happy
now
expect
quite
etc
use
daughter
better
magic
big
clean
recommend
long
little
area
go back
crowded
crowd
child
castle
seem
make sure
want
stayed
nice
closed
character
expensive
front
one day
fantastic
enjoyed
waiting
Hong Kong
stay
two
come
say
meet
busy
though
plenty
Florida
Small World
Disney World
Disneyland Park
drink
find
plan
fast passes
everything
year
old
love
great time
theme park
cast member
shop
Overall
disappointed
spent
bring
spend
way
early
adult
buy
u
got
Indiana Jone
Pirate
min
walk
age
ride
right
train
wait time
let
give
although
earth
Disney park
loved
many
least
sure
able
eat
friendly
first time
part
start
left
used
couple
getting
World
Main Street
Thunder Mountain
Ya
feels
small
must
said
especially
bit
enjoy
wanted
Florida
try
look
trip
first
price
restaurant
make
enough
come
two
stay
due
awesome
may
fantastic
one day
enjoyed
waiting
stayed
want
nice
closed
character
crowd
child
castle
seem
make sure
crowded
go back
area
long
little
people
visit
see
food
adult
buy
recommend
clean
helpful
magic
big
miss
daughter
feel
definitely
better
got
use
quite
etc
expect
now
wait
great
still
went
take
u
way
spend
bring
early
Overall
disappointed
shop
cast member
everything
year
old
love
great time
theme park
Indiana Jone
Pirate
min
walk
age

Text Preprocessing

1. Text cleaning

(lowercasing, puntuation/numbers/extra white spaces removal)

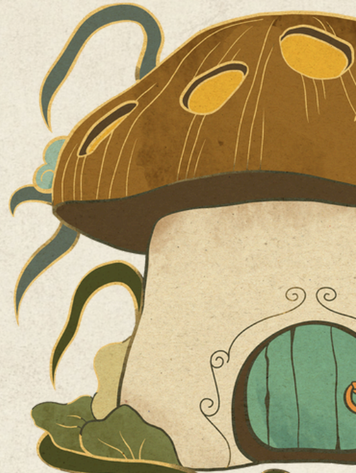
2.Tokenization

3. Lemmatization

4.Stopwords removal

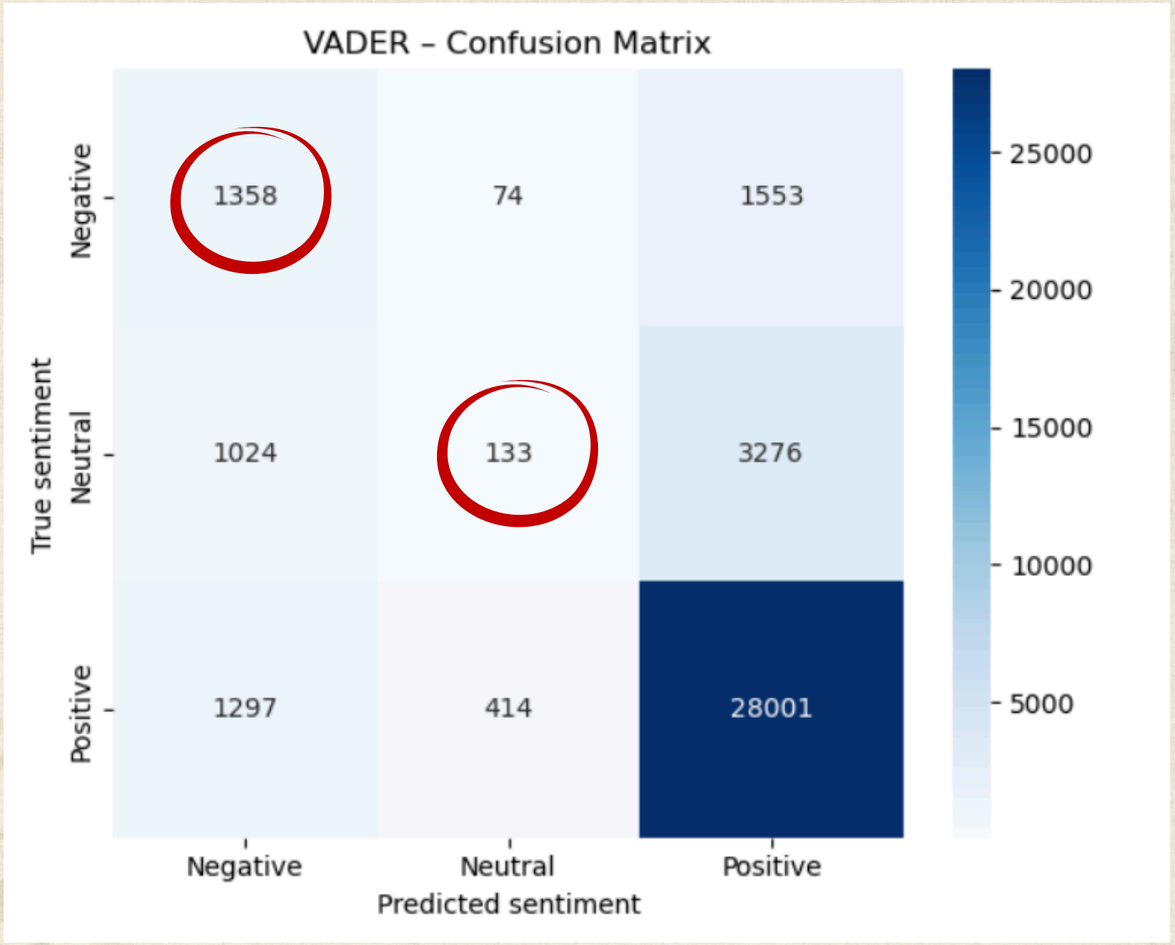
After cleaning we had 60k unique tokens and after lemmatization 56k.
Total number of tokes = 232 k

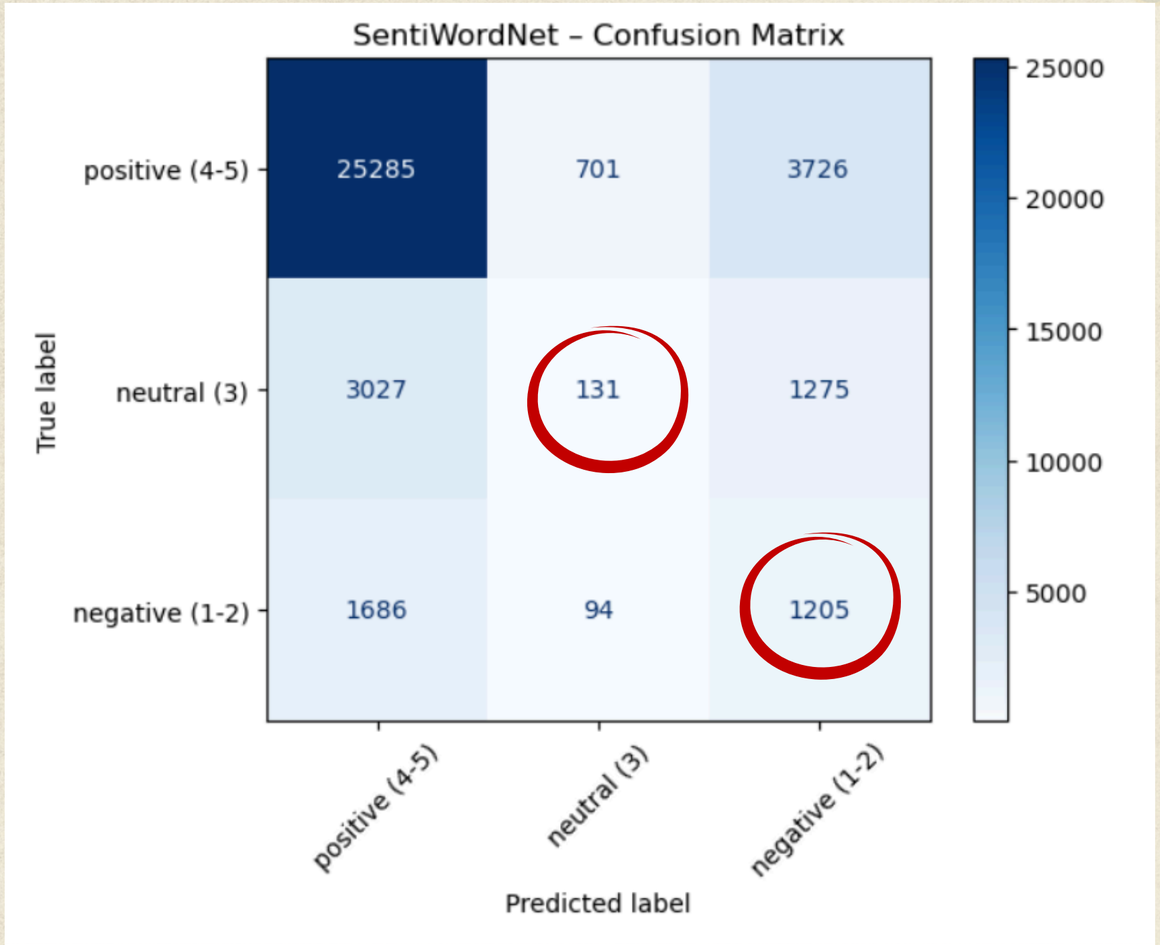
At the end we did TF – IDF computation



VADER vs SentiWordNet

Vader and SWN labels: score>= 0.05 - positive (4-5) , score <= −0.05 negative (1-2)





Accuracy 79,43 % F1 score 75,56%

Accuracy 71,70 % F1 score 70,46%

```python
def map_bert_to_rating(row):
    label = row["label"]
    score = row["score"]

    if label == "POSITIVE" and score >= 0.6:
        return "positive (4-5)"
    elif label == "NEGATIVE" and score >= 0.6:
        return "negative (1-2)"
    else:
        return "neutral (3)"
```

BERT

Accuracy 74.17 %

F1 score 74.48 %

	precision	recall	f1-score	support
negative (1-2)	0.26	0.92	0.41	2985
neutral (3)	0.18	0.02	0.04	4433
positive (4-5)	0.94	0.83	0.88	29712
accuracy			0.74	37130
macro avg	0.46	0.59	0.44	37130
weighted avg	0.80	0.74	0.74	37130

Machine Learning Models

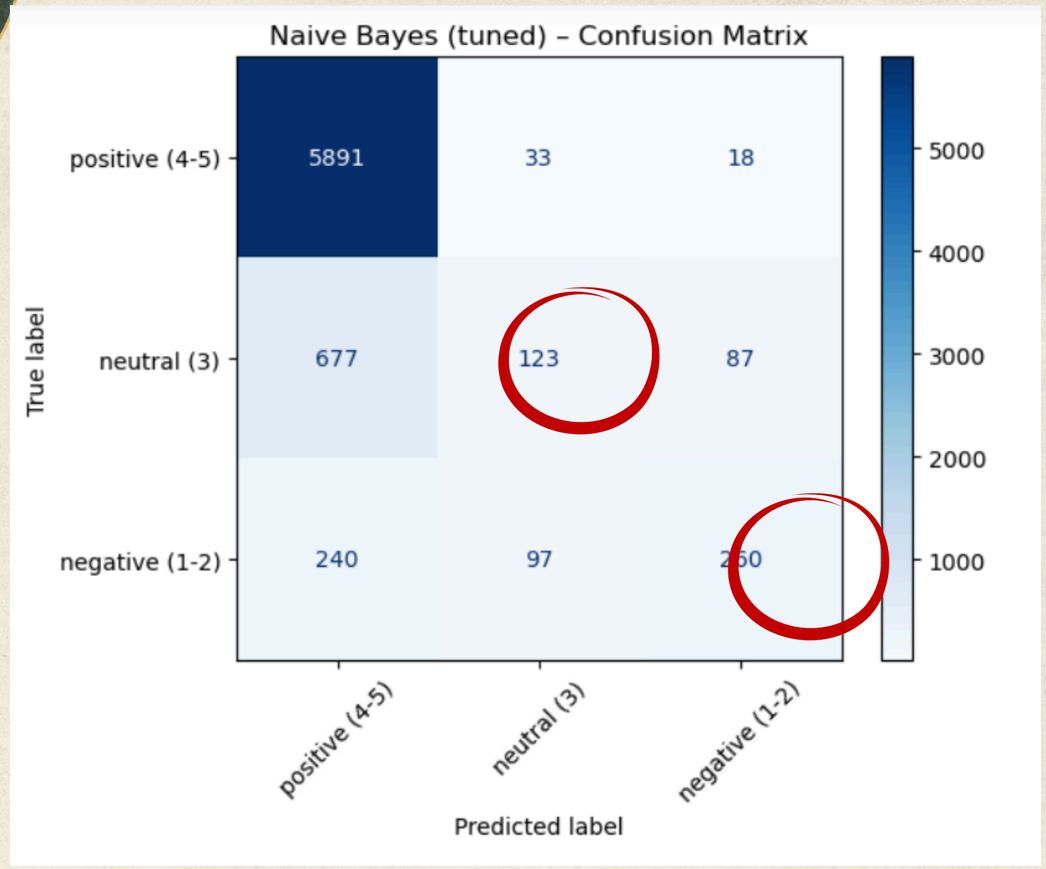
Train, test split 80/20, Tfidf Vectorizer (min_df=5, max_df=0.9), crossvalidation, hyperparameters tunning

- Multinominal Naive Bayes : alpha = 0.05
- Logistic regression
- KNN : 5 neighbors, Manhattan distance
- Random Forest: max depth =None, min samples leaf =1, min samples split =2, n estiamtors =200
- Linear SVC: C =1
- XGB: max depth =6, learning rate = 0.1, subsample =0.8, colsample by tree=0.8, n estimators= 300

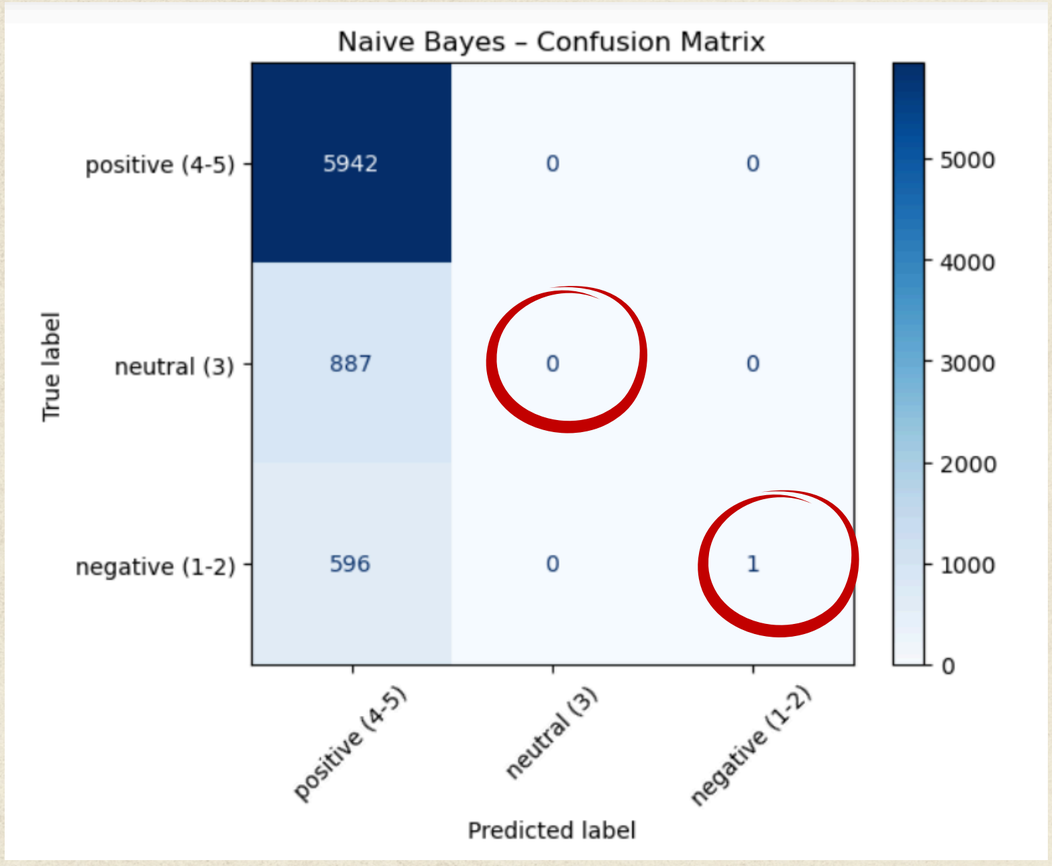
Machine Learning Models

Model	Accuracy	F1 score
Multinominal Naive Bayes	84.49%	80.86%
Logistic Regression	84.96%	81.93%
KNN	80.37%	76.17%
Linear SVC	84.78%	82.69%
Random Forest	80.11 %	71.36 %
XGB	84.28 %	80.91%

Tuned vs not tuned

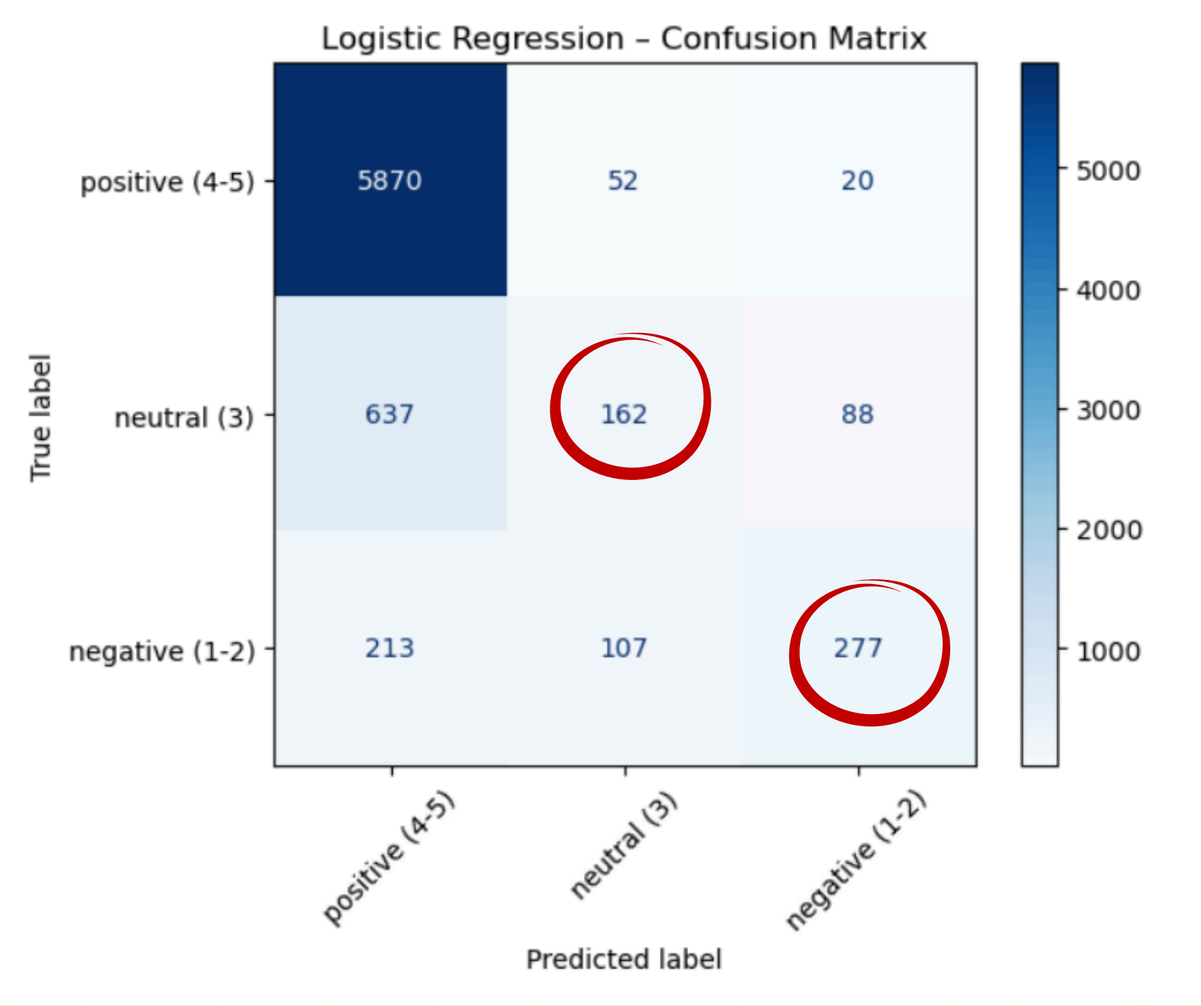


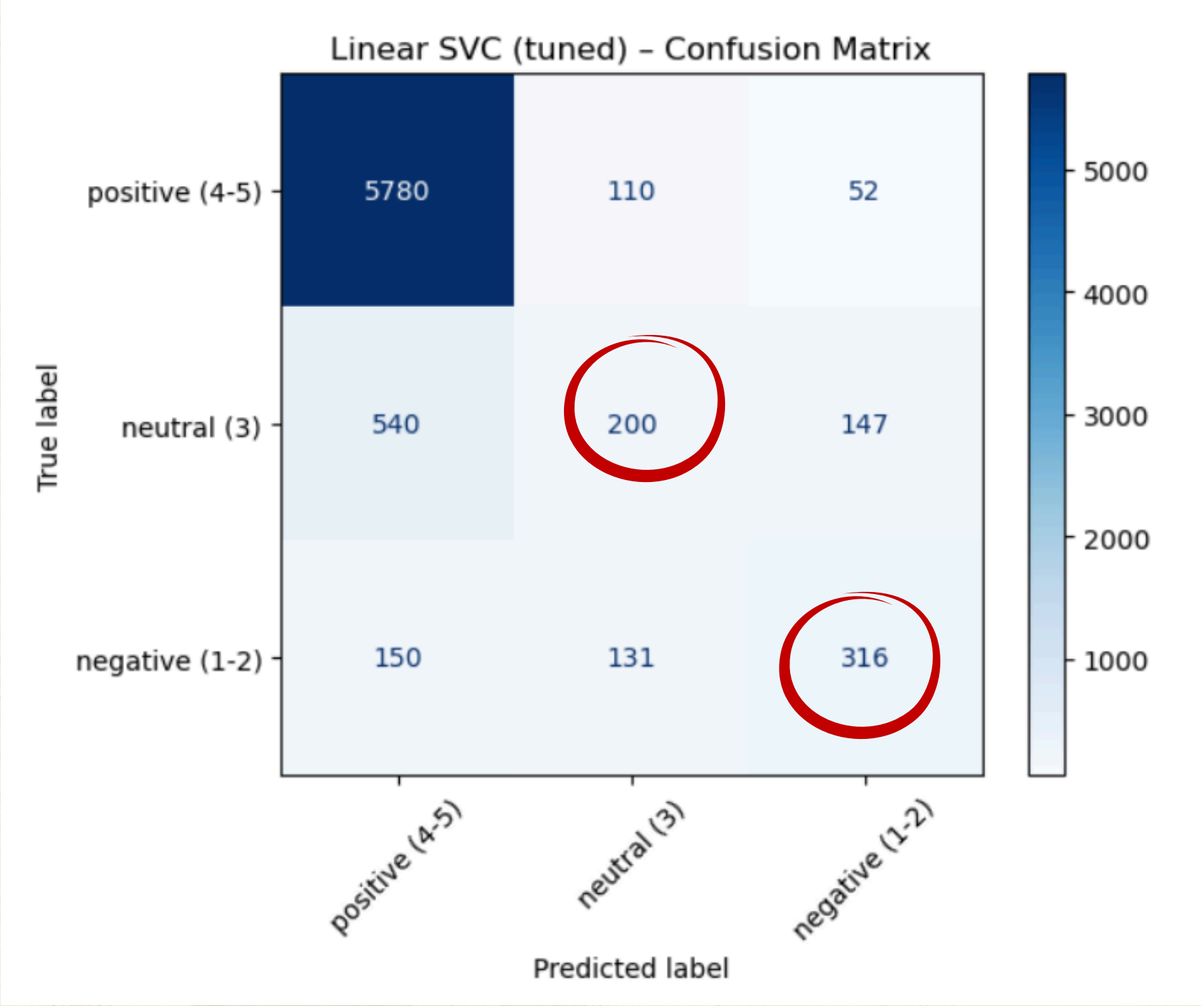
Accuracy 84,49% F1 score 80,86%



Accuracy 80.03% F1 score 71,17%

Logistic Regression vs Linear SVC





THANK YOU

See You In Disneyland